

Hong Kong Physics Olympiad 2025

2025 年香港物理奧林匹克競賽

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The Physical Society of Hong Kong

香港物理學會

Hong Kong Physics Olympiad Committee

香港物理奧林匹克委員會

18 May, 2025

2025 年 5 月 18 日

Rules and Regulations 競賽規則

1. All questions are in bilingual versions. You can answer in either Chinese or English, but only ONE language should be used throughout the whole paper.
所有題目均為中英對照。你可選擇以中文或英文作答，惟全卷必須以單一語言作答。
2. The multiple-choice answer sheet will be collected 1.5 hours after the start of the contest. You can start answering the open-ended questions any time after you have completed the multiple-choice questions without waiting for announcements.
選擇題的答題紙將於比賽後一小時三十分收回。若你在這之前已完成了選擇題，你亦可開始作答開放式題目，而無須等候任何宣佈。
3. On the cover of the answer book and the multiple-choice answer sheet, please write your 8-digit Contestant Number, English Name, and Seat Number.
在答題簿封面及選擇題答題紙上，請填上你的 8 位數字參賽者號碼、英文姓名、及座位號碼。
4. After you have made the choice in answering a multiple choice question, fill the corresponding circle on the multiple-choice answer sheet fully using a HB pencil.
選定選擇題的答案後，請將選擇題答題紙上相應的圓圈用 HB 鉛筆完全塗黑。
5. The open-ended problems are quite long. Please read the whole problem first before attempting to solve them. If there are parts that you cannot solve, you are allowed to treat the answer as a known answer to solve the other parts.
開放式問答題較長，請將整題閱讀完後再著手解題。若某些部分不會做，也可把它們的答案當作已知來做其他部分。
6. All numerical answers must be in (at least) 3 significant digits.
所有的數值答案必須保留至少 3 個有效數字。

The following symbols and constants are used throughout the examination paper unless otherwise specified:

除非特別注明，否則本卷將使用下列符號和常數：

Gravitational acceleration on Earth surface 地球表面重力加速度	g	9.80 m s^{-2}
Gravitational constant 重力常數	G	$6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$

There is no friction in the problems unless otherwise specified.

除非另有說明，否則問題中沒有摩擦力。

Multiple Choice Questions (2 Marks Each) 選擇題(每題 2 分)

1. Alice launches a particle from the surface of a uniform, stationary spherical planet at an angle to the vertical. The particle travels without air resistance and eventually falls back to the planet. Bob describes the path of the particle as a parabola, using the laws of projectile motion. Meanwhile, Alice recalls from Kepler's laws that every bound orbit around a point mass is an ellipse (or circle) and that the gravitational effect of a uniform sphere is equivalent to that of a point mass. Which of the following best explains the discrepancy between their observations?

1. Alice 從一個均勻、靜止的球形行星表面，以一個與沿垂直方向成一定角度的方向發射一個粒子。該粒子在沒有空氣阻力的情況下飛行，最終落回行星表面。Bob 運用拋物體運動的定律，認為粒子的軌跡是一條拋物線。與此同時，Alice 根據開普勒定律回想起：繞著一個點質量運行的所有束縛軌道都是橢圓（或圓形），而一個均勻球體的引力效應等效於一個點質量。以下哪一項最能解釋他們觀察之間的差異？

(A) Because the experiment takes place very close to the surface of the sphere, it is no longer valid to replace the sphere with a point mass. 由於實驗發生在球體表面附近，因此已不再適用以點質量代替球體的假設。

(B) Because the particle strikes the ground, it is not in orbit around the planet and therefore can follow a non-elliptical path. 因為粒子最終會撞擊地面，所以它不構成對行星的軌道運動，因而可以遵循非橢圓形的路徑。

(C) Alice overlooked the fact that motions around a point mass can also be parabolas or hyperbolas. Alice 忽略了繞點質量運動的軌跡也可以是拋物線或雙曲線的事實。

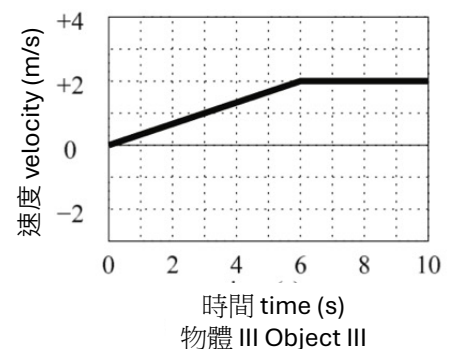
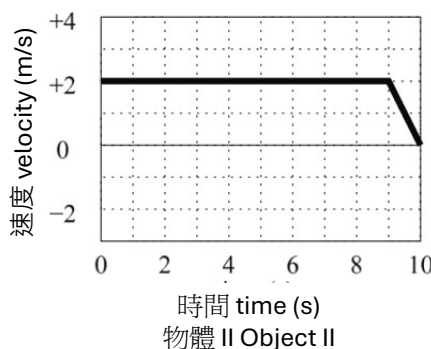
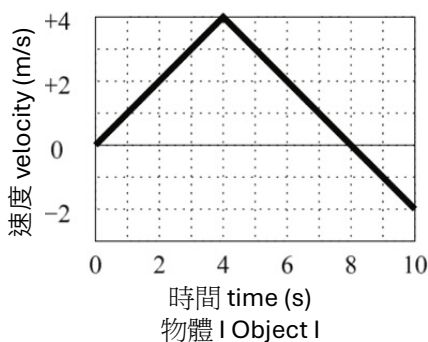
(D) Kepler's laws only apply in the limit of large orbits. 開普勒定律僅適用於軌道半徑很大的情況。

(E) The path is an ellipse, but it closely resembles a parabola due to the short distance of the flight relative to the distance from the center of the planet. 軌跡實際上是一條橢圓，但由於飛行距離相對於行星中心的距離很短，所以看起來像是一條拋物線。

Answer: E

Questions 2 to 4 refer to the three graphs below which show velocity of three objects as a function of time. Each object is moving only in one dimension.

第 2 至第 4 題參考下方的三張圖表，這些圖顯示三個物體的速度隨時間變化的關係。每個物體僅在一維方向上運動。



2 Rank the **magnitudes of the average acceleration** during the ten second interval.

2. 請排列三個物體在這十秒時間區間內平均加速度大小的順序。

(A) I>III>II

(B) II>I=III

(C) II>I>III

(D) I>II=III

(E) I=II=III

Answer: E

3. Rank the magnitude of the instantaneous acceleration at $t = 5$ s.

3. 請排列三個物體在 $t = 5$ s 時的瞬時加速度大小順序。

(A) I>III>II

(B) II>I=III

(C) II>I>III

(D) I>II=III

(E) I=II=III

Answer: A

For I, $|a| = 1 \text{ m/s}^2$ For II, $a = 0$ For III, $a = 0.5 \text{ m/s}^2$

4. Rank the magnitudes of the displacement travelled during the ten second interval.

4. 請排列三個物體在這十秒時間區間內位移大小的順序。

(A) I>III>II

(B) II>I=III

(C) II>I>III

(D) I>II=III

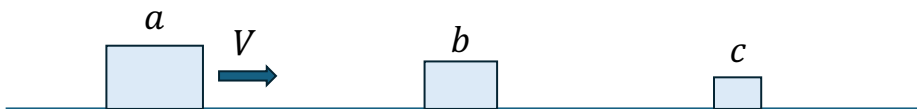
(E) I=II=III

Answer: B

For I, displacement = $16 - 2 = 14\text{m}$ For II, displacement = $18 + 1 = 19\text{m}$ For III, displacement = $6 + 8 = 14 \text{ m}$

5. As shown in figure, small blocks a , b , and c are at rest on a smooth horizontal surface. Now, let block a move to the right with a speed V and collide elastically with block b . Then, block b will also collide elastically with block c . If the masses of blocks b and c can be chosen arbitrarily, what is the maximum speed of block c after the first collision between blocks b and c ?

5. 如圖所示，小方塊 a 、 b 和 c 靜止在一個光滑的水平面上。現在，讓方塊 a 以速度 V 向右移動並與方塊 b 發生彈性碰撞。接著，方塊 b 也會與方塊 c 發生彈性碰撞。如果方塊 b 和方塊 c 的質量可以任意選擇，那麼在方塊 b 和方塊 c 的第一次碰撞之後，方塊 c 的最大速率是多少？

(A) V (B) $2V$ (C) $3V$ (D) $4V$ (E) $5V$

Solution: (D)

Let the masses of 3 block be a, b, c .(1) Collision between a and b . By the conservation of energy and momentum

$$aV = av_a + bv_b$$

$$aV^2 = av_a^2 + bv_b^2$$

$$a(V - v_a)(V + v_a) = bv_b^2 \Rightarrow V + v_a = v_b$$

$$\Rightarrow aV = a(v_b - V) + bv_b$$

$$\Rightarrow v_b = \frac{2aV}{a+b} \leq 2V$$

(2) Collision between b and c . Similarly, we can conclude that

$$v_c = \frac{2b}{b+c} v_b = \frac{4ab}{(a+b)(b+c)} V \leq 4V$$

6. As shown in Figure, object Q is connected to a massless spring and is initially at rest on a frictionless horizontal surface. Object P moves with a velocity v and collides elastically with the spring and object Q . It is known that the collision is completely elastic and that both objects have equal mass. During the collision process, under which of the following conditions is the spring at its maximum compression?

6. 如圖所示，物體 Q 連接在一條無質量的彈簧上，初始時靜止在一個無摩擦的水平面上。物體 P 以速度 v 運動，並與彈簧以及物體 Q 發生彈性碰撞。已知碰撞是完全彈性的，且兩個物體的質量相等。在碰撞過程中，下列哪一種情況發生時，彈簧的壓縮量最大？



- (A) When the velocity of P is exactly zero. 當 P 的速度正好為零時。
- (B) When the velocities of P and Q are equal. 當 P 和 Q 的速度相等時。
- (C) When Q just begins to move. 當 Q 剛開始移動時。
- (D) When the velocity of Q is equal to v . 當 Q 的速度等於 v 時。
- (E) When P has just transferred all its kinetic energy to the spring. 當 P 剛剛將所有的動能轉移到彈簧上時。

Answer: B

The spring has maximum compression when $\frac{dx}{dt} = 0$, where $x = x_Q - x_P$ is the compression of the spring.

$$\Rightarrow \frac{dx_Q}{dt} = \frac{dx_P}{dt}$$

Their velocities are equal.

7. A person P is running up a long escalator that is moving downward with a constant velocity v_e . The person P runs upward relative to the escalator with a speed v_p . There are two scenarios to consider:

Case 1: The person runs up the escalator for a time t_1 and reaches the top.

Case 2: If the escalator were not moving, the same person, running with the same speed v_p , would take a time t_2 to reach the top.

Which of the following equations correctly relates t_1 and t_2 ?

7. 一個人 P 正在跑上向下移動的長型扶手電梯，該扶手電梯以恆定速度 v_e 向下移動。此人 P 相對於扶手電梯以速度 v_p 向上跑。有兩種情況需要考慮：

情況 1：此人跑上扶手電梯並花費時間 t_1 到達頂端。

情況 2：如果扶手電梯不移動，這個人以相同的速度 v_p 跑，則他需要花費時間 t_2 到達頂端。

以下哪個方程式正確地描述 t_1 和 t_2 之間的關係？

- (A) $t_1 = t_2 + \frac{v_e}{v_p}$
- (B) $t_1 = \frac{t_2}{1 - \frac{v_e}{v_p}}$
- (C) $t_1 = t_2 \left(1 - \frac{v_e}{v_p}\right)$
- (D) $t_1 = t_2 \left(1 + \frac{v_e}{v_p}\right)$
- (E) $t_1 = t_2 - \frac{v_e}{v_p}$

Solution: B

Let the distance of the escalator be L .

$$t_1 = \frac{L}{v_p - v_e}, \quad t_2 = \frac{L}{v_p}$$

$$\Rightarrow t_1 = \frac{v_p}{v_p - v_e} t_2 = \frac{1}{1 - \frac{v_e}{v_p}} t_2$$

8. A block, X, is travelling at speed v_0 toward another block, Y, which is initially at rest. They undergo head-on elastic collision. Then block X bounces backward the direction where it comes from. The collision occurs on a horizontal smooth surface. Which of the following statement is correct?

8. 有一個物件 X 以速度 v_0 朝向另一個物件 Y 移動，物件 Y 最初靜止。它們進行正面彈性碰撞，然後物件 X 反彈回來的方向。碰撞發生在水平光滑表面上。以下哪個敘述是正確的？

(A) Block X and block Y have the same mass. 物件 X 和物件 Y 的質量相同。

(B) Block X is heavier than block Y. 物件 X 比物件 Y 重。

(C) Block X is lighter than block Y. 物件 X 比物件 Y 輕。

(D) If the initial speed of block X is higher than v_0 , then block X will not bounce backward. 如果物件 X 的初始速度高於 v_0 ，則物件 X 不會反彈回來。

(E) If the initial speed of block X is lower than v_0 , then block X will not bounce backward. 如果物件 X 的初始速度低於 v_0 ，則物件 X 不會反彈回來。

Answer: C

9. Consider an alien planet, A, far away from us. Planet A is not rotating and there are no suns (stars) near Planet A. The alien on the planet created an artificial sun that orbits in a circle around Planet A such that there are 24 hours (one day on Earth) between each artificial sunrise. Despite the question whether this is a smart way to create one day on the planet, find the radius of the circular orbit of the artificial sun around Planet A.

Given the mass of Planet A is 10^{24} kg and the mass of the artificial sun is 10^{14} kg

9. 考慮一個距離我們很遠的外星行星 A。行星 A 不旋轉，附近沒有太陽（恒星）。上面的外星人創造了一個環繞行星 A 的人工太陽，使得每次人工日出之間有 24 小時（地球上的一天）。不論這是不是一個聰明的方法來創造行星上的一天，找出人工太陽環繞行星 A 的圓形軌道半徑。

已知行星 A 的質量為 10^{24} kg，人工太陽的質量為 10^{14} kg。

(A) 20.7 km

(B) 256,000 km

(C) 10.8 km

(D) 12.6×10^{19} km

(E) 23,300 km

Answer: E

Solution:

Kelper's 3rd Law

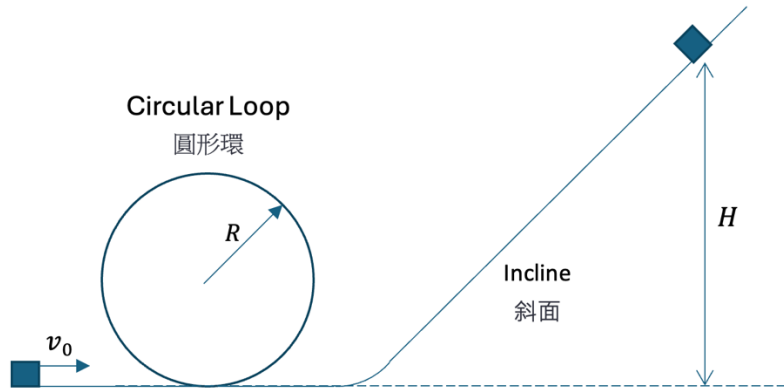
$$r^3 = \frac{GM}{4\pi^2} T^2$$

Where M is the mass of planet A and T is 24 hours.

10. A small block is moving at speed v_0 on a track that includes a circular loop and an incline. The track is frictionless. There is no attachment/mounting mechanism between the track and the block. The block completes the loop without falling and then reach a vertical height H up the incline. Find the minimum value of H that the block can reach on the incline after completing the loop.

10. 一個小方塊以速度 v_0 在包含一個圓形環和一個斜面的軌道上移動。軌道是無摩擦力的。方塊與軌道之間沒有任何附著或安裝機制。方塊在不掉落的情況下完成環形軌道，然後在斜面上達到垂直高度 H 。

找出方塊在完成環形軌道後可以在斜面上達到的最小 H 值。



- (A) $2R$ (B) $5R$ (C) $\frac{5}{2}R$ (D) $\frac{1}{2}R$ (E) $\frac{v_0^2}{2g} + 2R$

Answer: C

Solution: The minimum speed at the top of the loop is $v_{top} = \sqrt{gR}$. Thus, when the block traveled to a height $2R$ on the incline, it has the speed $\sqrt{gR}$ which can be further converted into potential energy of $mg\Delta H = \frac{1}{2}mv_{top}^2 = \frac{1}{2}mgR$.

Therefore, it can further travel $\Delta H = \frac{R}{2}$ from $2R$ along the incline. So $H = 2R + \frac{1}{2}R = \frac{5}{2}R$

11. Two race cars are initially at rest. They start moving forward at $t = 0$ s with different but constant accelerations. One of the car is 1 m ahead of the other one at $t = 1$ s. At what time will they be 10 m away from each other if their individual accelerations are constant?

11. 兩輛賽車最初靜止。它們在 $t = 0$ s 開始向前移動，並具有不同但恆定的加速度。在 $t = 1$ s 時，其中一輛車比另一輛車領先 1 m。假設它們的個別加速度是恆定的，問它們在什麼時刻會相距 10 m？

- (A) 1.8 s (B) 3.2 s (C) 5 s (D) 10 s (E) 20 s

Answer: B

Solution:

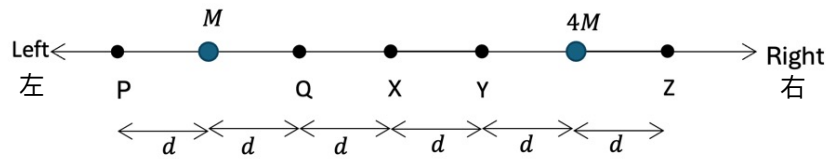
The distance of the cars is $\Delta x = \frac{1}{2}(a_1 - a_2)t^2 \propto t^2$.

The time scaled with the distance as $t \propto \sqrt{\Delta x}$

So the time when they are 10 m apart = $t_{10} = t_1 * \sqrt{\frac{10}{1}} = 3.16 \text{ s} \approx 3.2 \text{ s}$

12. Two planets with different masses, M and $4M$, are separated by a distance of $4d$ as shown in the figure below. Which of the following indicate the correct direction of the net gravitational force due to both planets at the points P, Q, X, Y and Z in the figure. Use Newton's Law of gravity in this problem.

12. 兩顆質量不同的行星，質量分別為 M 和 $4M$ ，彼此間隔距離為 $4d$ ，如下圖所示。以下哪個選項正確指示了在點 P、Q、X、Y 和 Z 處，由於兩顆行星引力而來的合力方向？在此問題中使用牛頓的萬有引力定律。



	P	Q	X	Y	Z
(A)	Right 右	Right 右	Right 右	Right 右	Left 左
(B)	Right 右	Left 左	Left 左	Left 左	Left 左
(C)	Right 右	Left 左	Zero Force 零力	Right 右	Left 左
(D)	Right 右	Left 左	Left 左	Right 右	Left 左
(E)	Right 右	Left 左	Right 右	Right 右	Left 左

Answer: E

Solution: Suppose the point with zero gravity is r to the right away from M. It can be solve from

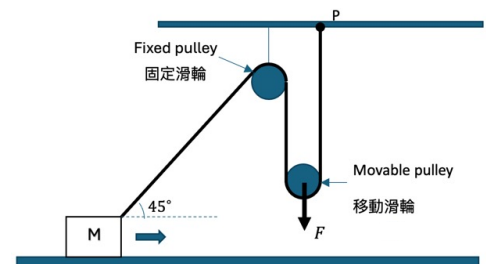
$$\frac{M}{r^2} = \frac{4M}{(4d - r)^2}$$

$$r = \frac{4}{3}d$$

Therefore, it is between Q and X. The gravity from M is larger than the gravity from $4M$ at Q. The gravity from $4M$ is larger than the gravity from M at X and Y.

13. A block of mass $M = 1 \text{ kg}$ is being pulled with a rope. One end of the rope is fixed on the ceiling at point P. The rope goes through two pulleys that one is fixed from the ceiling and the other one is movable vertically as shown in the diagram below.

13. 一個質量為 $M = 1 \text{ kg}$ 的物件正在被一根繩子拉動。繩子的一端固定在天花板上的點 P。繩子穿過兩個滑輪，其中一個固定在天花板上，另一個可以垂直移動，如下圖所示。



At the moment shown in the diagram, a force F is exerted downward on the movable pulley and the block is accelerating to the right at 4ms^{-2} . Assuming there is no friction between the block and the ground, the pulleys and the rope are massless and the rope does not slip on the pulleys. Find the value of F at the moment shown in the diagram.

在圖中所示的時刻，一個力 F 作用在可移動滑輪上向下拉，物件正以 4ms^{-2} 的加速度向右運動。假設物件與地面之間沒有摩擦，滑輪和繩子是無質量的，且繩子在滑輪上不滑動。找出圖中所示時刻的力 F 的值。

- (A) 4.0 N (B) 8.0 N (C) 11 N (D) 16 N (E) 5.7 N

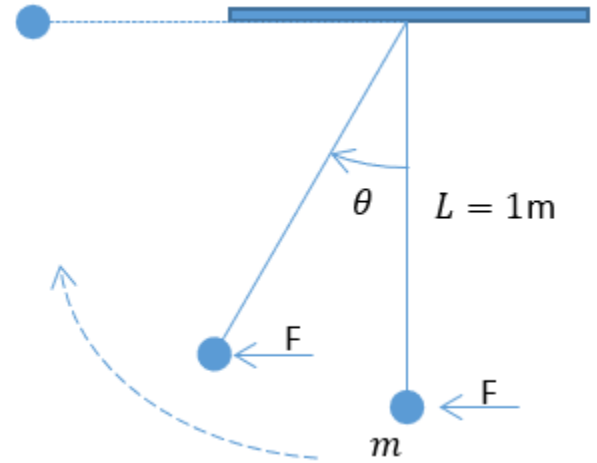
Answer: C

Solution: The tension of the rope is $T = \frac{Ma}{\sin 45^\circ} = 5.657 \text{ N}$. Applying Newton's second law at the massless movable pulley, we have

$$F = 2T = 11.3 \text{ N}$$

14. A simple pendulum with mass $m = 3 \text{ kg}$ and length $L = 1 \text{ m}$ is initially at rest in the vertical position. Let θ be the angle between the rope of the pendulum to the vertical. A constant horizontal force F is exerted on the mass from $\theta = 0^\circ$ to $\theta = 30^\circ$. The force is then removed. If the angle of the pendulum can reach $\theta = 90^\circ$ after the force is removed, find the minimum value of F .

14. 一個簡單的單擺，其質量為 $m = 3 \text{ kg}$ ，長度為 $L = 1 \text{ m}$ ，最初靜止在垂直位置。設 θ 為單擺繩子與垂直方向之間的角度。一個恆定的水平力 F 從 $\theta = 0^\circ$ 作用單擺的在質量上直到 $\theta = 30^\circ$ 。然後該力被移除。如果在力移除後，單擺的角度可以達到 $\theta = 90^\circ$ ，找出 F 的最小值。



- (A) 33.9 N (B) 29.4 N (C) 46.9 N (D) 63.1 N (E) 58.8 N

Answer: E

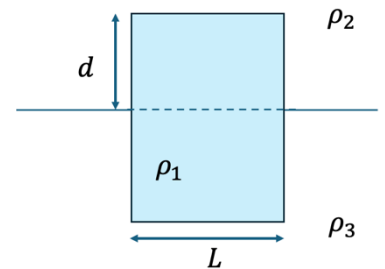
Solution: The work done by the force is $W = F * L * \sin 30^\circ$.

According to work-energy theorem,

$$W = \Delta E = \Delta U = mgL$$

$$F = \frac{mg}{\sin 30^\circ} = 58.8 \text{ N}$$

15. A container is separated into the upper and lower compartment by a partition with the upper compartment containing a liquid with density ρ_2 and the lower compartment containing a liquid with density ρ_3 . The partition has a square hole with side length L . A cube with side length L and density ρ_1 can remain in stable equilibrium under gravity with its top face at a distance d from the partition, where $0 < d < L$. The surface of the cube and the edges of the hole are smooth. The liquids are also non-viscous. Which of the following statements about the densities is/are true?



15. 一個容器被隔板分隔為上層與下層隔室，上層隔室裝有密度為 ρ_2 的液體，下層隔室裝有密度為 ρ_3 的液體。隔板上有一個邊長為 L 的正方形孔洞。一個邊長為 L 、密度為 ρ_1 的立方體能在重力作用下保持穩定平衡，使得其頂面與隔板保持距離 d （其中 $0 < d < L$ ）。立方體表面與孔洞邊緣皆光滑，液體亦為無粘性。下列關於密度的敘述何者正確？

- I. $\rho_1 > \rho_2 > \rho_3$ II. $\rho_2 > \rho_1 > \rho_3$ III. $\rho_3 > \rho_2 > \rho_1$ IV. $\rho_3 > \rho_1 > \rho_2$
 (A) I only 只有 I (B) II only 只有 II (C) III only 只有 III
 (D) IV only 只有 IV (E) I and IV I 和 IV

Answer: D

$$\begin{aligned} F(x) &= \rho_2 g L^2 x + \rho_3 g L^2 (L - x) - \rho_1 g L^3 \\ &= (\rho_3 - \rho_1) g L^3 + (\rho_2 - \rho_3) g L^2 x \end{aligned}$$

$$F(d) = 0 \Rightarrow d = \frac{\rho_1 - \rho_3}{\rho_2 - \rho_3} L$$

$$F(x) = (\rho_2 - \rho_3)gL^2(x - d)$$

$$d > 0 \Rightarrow \rho_1, \rho_2 > \rho_3 \text{ OR } \rho_1, \rho_2 < \rho_3$$

But stable equilibrium $\rightarrow \rho_2 < \rho_3$, so $\rho_1, \rho_2 < \rho_3$

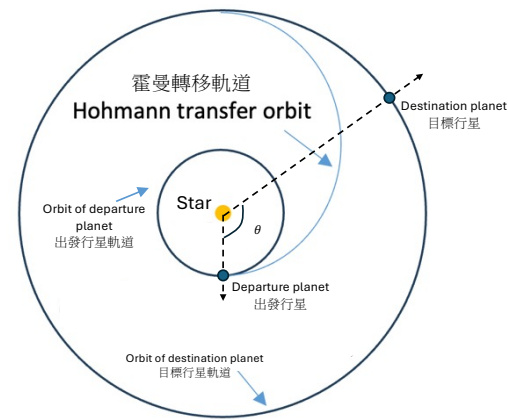
$$d < L \Rightarrow \frac{\rho_1 - \rho_3}{\rho_2 - \rho_3} < 1$$

$$\Rightarrow \rho_1 - \rho_3 > \rho_2 - \rho_3$$

$$\Rightarrow \rho_1 > \rho_2$$

In conclusion, $\rho_3 > \rho_1 > \rho_2$.

16. The most efficient way to send a spacecraft from one planet to another is by using a Hohmann transfer orbit, as shown in the figure below. If the orbits of the departure and destination planets are both circular, the Hohmann transfer orbit is an elliptical orbit whose perihelion and aphelion are tangent to the orbits of the two planets. The rockets are fired briefly at the departure planet to put the spacecraft into the transfer orbit; the spacecraft then coasts until it reaches the destination planet. The rockets are then briefly fired again to put the spacecraft into the same orbit about the star as the destination planet. Assume that the orbital radius of the destination planet is 3 times that of the departure planet and the two orbits are in the same direction. To reach the destination planet from the departure planet, the launch must be timed so that the destination planet will be at the right spot when the spacecraft reaches its orbit around the star. At launch, what must the angle θ between the star-destination planet line and the star-departure planet line be?



16. 將太空船從一個行星轉移到另一個行星的最有效方法是使用如下圖所示的霍曼轉移軌道。如果出發行星和目標行星的軌道均為圓形，則霍曼轉移軌道是一個橢圓軌道，其近日點和遠日點分別與兩行星的軌道相切。火箭在出發行星短暫點火，將太空船送入轉移軌道；隨後太空船無動力運行直至抵達目標行星。此時火箭再次短暫點火，使太空船進入與目標行星相同的繞恆星軌道。假設目標行星的軌道半徑為出發行星的 3 倍，且兩軌道運行方向相同。要從出發行星抵達目標行星，必須選擇適當的發射時機，使太空船到達目標軌道時，目標行星恰好位於相應位置。求發射時，恆星-目標行星連線與恆星-出發行星連線之間必須形成何夾角 θ ？

- (A) 27° (B) 53° (C) 82° (D) 114° (E) 146°

Answer: C

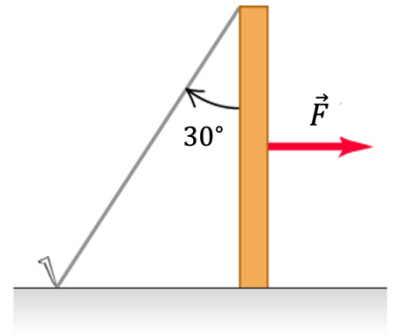
The semi-major axis of the transfer orbit is $\frac{1+3}{2} = 2$. According to the Kepler's 3rd law,

$$T = \sqrt{2^3} \Rightarrow \frac{T}{2} = \sqrt{2}$$

$$T' = \sqrt{3^3}$$

$$180^\circ - \frac{\sqrt{2}}{\sqrt{3^3}} \times 360^\circ = 82^\circ$$

17. One end of a post weighing W and with height H rests on a rough horizontal surface with coefficient of static friction μ_s . The upper end is held by a rope fastened to the surface and making an angle of 30° with the post, as shown in the figure below. A horizontal force F to the right is exerted on the post. If the point of application of the force is too high, the post cannot be made to slip, no matter how great F is. Find the critical height for the point of application.



17. 一根重量為 W 、高度為 H 的柱子，其一端置於靜摩擦係數為 μ_s 的粗糙水平面上。柱子上端由一根固定於該平面並與柱子成 30° 夾角的繩索牽引，如下圖所示。現對柱子施加向右的水平力 F 。若施力點位置過高，則無論 F 多大，皆無法使柱子滑動。求此施力點的臨界高度。

- (A) $\frac{H}{1+\sqrt{3}\mu_s}$ (B) $\frac{\sqrt{3}H}{\sqrt{3}+\mu_s}$ (C) $\frac{2H}{2+\sqrt{3}\mu_s}$ (D) $\frac{H}{1+2\mu_s}$ (E) $\frac{\sqrt{3}H}{\sqrt{3}+2\mu_s}$

Answer: A

$$F = T \sin \theta + f$$

$$N = T \cos \theta + W$$

$$TH \sin \theta = Fh$$

$$F = \frac{Fh}{H} + f$$

$$f = \left(1 - \frac{h}{H}\right)F \geq 0$$

$$\mu_s N = \mu_s (T \cos \theta + W) = \mu_s \left(\frac{Fh}{H} \cot \theta + W\right)$$

$$f \leq \mu_s N \Rightarrow \frac{1 - \mu_s \frac{W}{F}}{1 + \mu_s \cot \theta} \leq \frac{h}{H}$$

18. Inside a fluid, an object with mass m starts from rest to move downward vertically under Earth's gravity and the frictional force of the fluid, of which the magnitude is given by

18. 在流體內部，一質量為 m 的物體受地球重力及流體摩擦力作用下從靜止開始鉛直向下運動，其中摩擦力大小由

$$f = \alpha v + \beta v^2,$$

where v is the speed of the object and $\alpha, \beta > 0$. What is the terminal velocity (the speed when time tends to infinity) of the object?

給出 (v 為物體速率, $\alpha, \beta > 0$)。物體的終端速度(當時間趨向無窮大時之速率)為何?

- (A) $\sqrt{\left(\frac{\alpha}{2\beta}\right)^2 + \frac{mg}{\beta}} - \frac{\alpha}{2\beta}$ (B) $\sqrt{\left(\frac{\alpha}{\beta}\right)^2 + \frac{mg}{\beta}} - \frac{\alpha}{\beta}$ (C) $\sqrt{\left(\frac{\alpha}{\beta}\right)^2 + \frac{mg}{\beta}} - \frac{\alpha}{2\beta}$
 (D) $\sqrt{\left(\frac{\alpha}{2\beta}\right)^2 + \frac{mg}{\beta}} - \frac{\alpha}{\beta}$ (E) $\sqrt{\left(\frac{\alpha}{2\beta}\right)^2 + \frac{mg}{\beta}} - \frac{2\alpha}{\beta}$

Answer: A

$$F = -\alpha v - \beta v^2 + mg = 0$$

$$v = \frac{\alpha \pm \sqrt{\alpha^2 + 4\beta mg}}{-2\beta} = \frac{-\alpha \mp \sqrt{\alpha^2 + 4\beta mg}}{2\beta} = \frac{\sqrt{\alpha^2 + 4\beta mg} - \alpha}{2\beta}$$

19. The upper end of a light rope of length L is attached to the ceiling, and a small steel ball with mass m is suspended from the lower end of the rope. Initially the ball is at rest and the rope is vertical. Then a constant horizontal force with magnitude F is applied and the ball moves in an arc of a circle of radius L . What is the speed of the ball when the rope makes an angle θ with the vertical?

19. 一條長度為 L 的輕質繩索上端固定於天花板，下端懸掛質量為 m 的小鋼球。初始時鋼球靜止且繩索呈鉛直狀態。隨後對鋼球施加大小恆為 F 的水平力，使其沿半徑 L 的圓弧運動。當繩索與鉛直方向夾角為 θ 時，鋼球的速率為何？

(A) $\sqrt{gL\left(\frac{F}{mg}\cos\theta + \sin\theta - 1\right)}$

(B) $\sqrt{2gL\left(\frac{F}{mg}\sin\theta + \cos\theta - 1\right)}$

(C) $\sqrt{2gL\left(\frac{F}{mg}\sin\theta + \cos\theta\right)}$

(D) $\sqrt{2gL\left(\frac{F}{mg}\sin\theta - \cos\theta\right)}$

(E) $\sqrt{gL\left(\frac{F}{mg}\sin\theta + \cos\theta - 1\right)}$

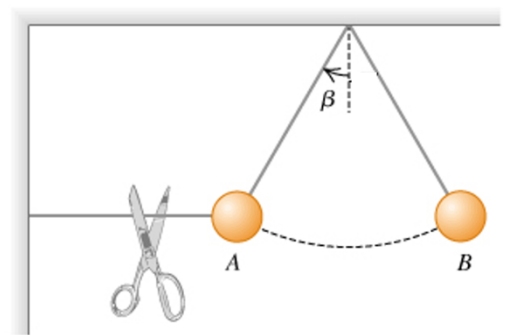
Answer: B

$$0 + 0 + 0 = \frac{1}{2}mv^2 + mgL(1 - \cos\theta) - FL\sin\theta$$

$$v = \sqrt{2gL\left(\frac{F}{mg}\sin\theta + \cos\theta - 1\right)}$$

20. A ball is held at rest at position A in the figure below by two light strings, with one being horizontal and the other making an angle of β with the vertical direction. The horizontal string is cut and the ball starts swinging as a pendulum under the Earth's gravity. Point B is the farthest to the right the ball goes as it swings back and forth. What is the ratio of the tension in the supporting string in position B to its value at A before the horizontal string was cut?

20. 如圖所示，一顆小球由兩根輕質細繩固定靜止於位置 A ，其中一繩呈水平，另一繩與鉛直方向夾角為 β 。當水平細繩被切斷後，小球開始在地球重力作用下以單擺形式擺動，點 B 為其擺動過程中最右側位置。求支撐細繩在位置 B 處的張力與水平繩切斷前在位置 A 處其張力之比值為何？」



A. $\frac{T_B}{T_A} = \sec^2\beta$

B. $\frac{T_B}{T_A} = \cos\beta$

C. $\frac{T_B}{T_A} = \sec\beta$

D. $\frac{T_B}{T_A} = \cos^2\beta$

E. $\frac{T_B}{T_A} = 1$

Answer: D

$$T_A \cos\beta = mg$$

$$T_B = mg \cos\beta$$

$$\frac{T_B}{T_A} = \cos^2\beta$$

Open Problems (20 Points Each) 開放題(每題 20 分)

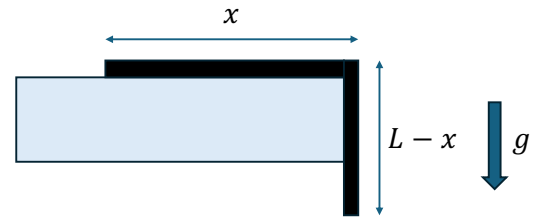
Hint: 提示:

Consider a force acting on an object of the form, $F(x) = kx$, where x is the displacement of the object. The work done by this force to move a particle by $x = a$ to $x = b$ is given by

考慮一種作用於物體的力，其可表達為 $F(x) = kx$ ，其中 x 是該物體位移。這個力對物體從 $x = a$ 移動到 $x = b$ 所做的功為

$$W_{a \rightarrow b} = \frac{k}{2}(b^2 - a^2)$$

1. [20] A rope is placed over the edge of a table, as shown in the diagram. The rope has a uniform linear mass density ρ , a total length L , and is subject to gravitational acceleration g . Assume that the initial length of the rope on the table is x_0 , and the rope is initially at rest, meaning $x(t = 0) = x_0$ and the initial velocity, $v(t = 0) = 0$. The effect of the rope bending over the edge is neglected (i.e., assume there is no friction at the edge and that the rope moves over a frictionless pulley).



1. [20] 一條繩子如圖所示地放在桌邊。該繩子具有均勻的線質量密度 ρ ，總長為 L ，並受到重力加速度 g 的作用。假設繩子起初在桌面上的長度為 x_0 ，且繩子最初靜止不動，也就是說 $x(t = 0) = x_0$ ，初速度 $v(t = 0) = 0$ 。忽略繩子在桌邊彎曲的影響（假設桌邊沒有摩擦，或可視為繩子在一個無摩擦的滑輪上運動）。

(a) [4] Assuming there is no friction between the table's surface and the rope, express the acceleration a of the slipping rope as a function of the length x of the rope remaining on the table.

(a) [4] 假設桌面與繩子之間沒有摩擦力，請將繩子滑落時的加速度 a 表示為繩子仍留在桌面上的長度 x 的函數。

(b) [5] Determine the velocity of the rope as a function of the remaining length x of the rope on the table.

(b) [5] 求繩子的速度 v 作為桌面上剩餘繩子長度 x 的函數。

(c) [5] Now, consider that there is static friction between the table's surface and the rope, with a coefficient of friction μ . Determine the length x_m at which the rope starts to slip as x decreases. Express the answer in terms of L, μ .

(c) [5] 現在考慮桌面與繩子之間存在靜摩擦力，摩擦係數為 μ 。求當繩子開始滑動時對應的長度 x_m ，也就是隨著 x 變小，繩子開始滑動時的 x 值。答案可用 L, μ 表達。

(d) [6] Given that the initial length of rope on the table is $x_0 = x_m$ and the kinetic friction also exists between the table and the rope, with the same coefficient of friction μ , determine the velocity of the rope at the moment when it completely leaves the table, i.e. find the velocity v when $x = 0$. Express the answer in terms of x_0, L, g, ρ, μ .

(d) [6] 已知繩子留在桌面上的初始長度為 $x_0 = x_m$ ，且桌面與繩子之間存在動摩擦力（摩擦係數同為 μ ），求出當繩子完全離開桌面那一刻的繩子速度，即求在 $x = 0$ 時的速度 v 。答案可用 x_0, L, g, ρ, μ 表達。

Solution:

(a)

$$\begin{aligned}\rho L a &= (L - x)\rho g \\ \Rightarrow a &= \left(1 - \frac{x}{L}\right)g\end{aligned}$$

(b) The initial PE is $U(x = L) = -(L - x_0)\rho g \times \frac{1}{2}(L - x_0) = -\frac{1}{2}\rho g(L - x_0)^2$

During the sliding of the rope, we have

$$U(x) = -\frac{1}{2}\rho g(L - x)^2$$

The conservation of mechanical energy gives,

$$\begin{aligned}\frac{1}{2}L\rho v^2 - \frac{1}{2}(L - x)^2\rho g &= -\frac{1}{2}(L - x_0)^2\rho g \\ \Rightarrow v(x) &= \sqrt{\frac{g}{L}((L - x)^2 - (L - x_0)^2)}\end{aligned}$$

(c) **(Force diagram)**

When the rope remains at rest due to the static friction, the normal force of the rope on the table is $N = x\rho g$.

The rope will slide when

$$\begin{aligned}(L - x)\rho g &\geq \mu x\rho g \\ \Rightarrow x_m &= \frac{L}{1 + \mu}\end{aligned}$$

(d) The normal force acting on the rope on the table is

$$N = \rho x g$$

And the kinetic friction is

$$f_k = \mu N = \mu \rho g x$$

The energy work theorem gives

$$\begin{aligned}-\frac{1}{2}(L - x_m)^2\rho g - \frac{1}{2}L\rho v^2 + \frac{1}{2}L^2\rho g &= \int_{x_m}^0 f_k dx = \frac{1}{2}\mu \rho g x_m^2 \\ \Rightarrow -\left(\frac{\mu}{1 + \mu}\right)^2 Lg - v^2 + Lg &= \frac{\mu g x_m^2}{L} = \frac{\mu g L}{(1 + \mu)^2} \\ \Rightarrow v^2 &= Lg \left(1 - \frac{\mu^2 + \mu}{(1 + \mu)^2}\right) = Lg \left(1 - \frac{\mu}{1 + \mu}\right) = \frac{Lg}{1 + \mu} \\ \Rightarrow v &= \sqrt{\frac{Lg}{1 + \mu}}\end{aligned}$$

2. [20] A basket trolley with mass $M = 3.0$ kg is initially at rest on a horizontal surface. The trolley is free to move (frictionless) along the x-axis, but it will not move vertically.

A particle with mass $m = 1.0$ kg flies into a trolley from the top left corner at angle θ and at speed $|\vec{v}| = v_0$ as shown in Figure 3(b). The particle then bounces off the walls in the basket with elastic collisions.

2. [20] 一個質量為 $M = 3.0$ kg 的購物籃手推車最初靜止在水平面上。手推車可以沿著 x 軸自由移動（無摩擦），但不會垂直移動。一個質量為 $m = 1.0$ kg 的粒子從左上角以角度 θ 和速度 $|\vec{v}| = v_0$ 飛入手推車，如圖 3(b) 所示。然後粒子在籃子裡的牆壁上進行彈性碰撞。

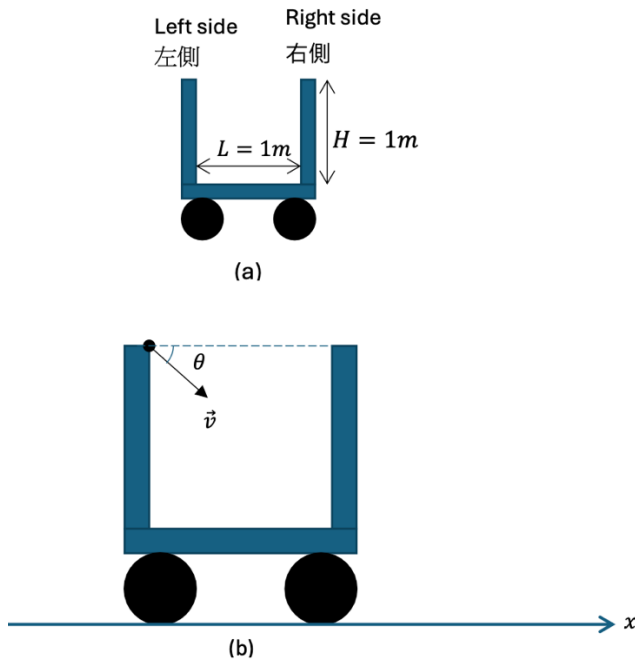


Figure 3: (a) Dimension of the trolley basket. (a) 手推車籃的尺寸。

(b) A particle enters the basket at the top left corner with a velocity $\vec{v}$.

(b) 一個粒子以速度矢量 $\vec{v}$ 從左上角進入籃子。

Let's first ignore the gravity acting on the particle for now. (Part (a) – (f))

Suppose the incident angle θ is small enough that the ball will at least hit the right side of the basket once.

讓我們先忽略作用在粒子上的重力。(部分 (a) – (f)) 假設入射角 θ 足夠小，使得球至少會碰到籃子右側一次。

(a) [2] Find the velocity along x-axis of the trolley with respect to the ground immediately after the ball hits the right side of the basket for the first time. Answer in terms of v_0 and θ . Take the right side as positive.

(a) [2] 找出球第一次碰到籃子右側後，手推車相對於地面沿 x 軸的速度。答案以 v_0 和 θ 表示。取右側為正。

(b) [2] Find the velocity along x-axis of the particle with respect to the trolley immediately after the ball hits the right side of the basket for the first time. Answer in terms of v_0 and θ . Take the right side as positive.

(b) [2] 找出球第一次碰到籃子右側後，粒子相對於手推車沿 x 軸的速度。答案以 v_0 和 θ 表示。取右側為正。

(c) [2] Suppose the particle will fly over the right side of the basket after at least one bounces off the side wall in the basket. Draw a diagram to indicate a trajectory of the particle with respect to the trolley (from the point of view in the co-moving reference frame of the trolley). There are multiple possible cases under the same condition. Draw only one of them.

(c) [2] 假設粒子在籃子側壁至少反彈一次後會飛越籃子的右側。畫一個圖示以表示粒子相對於手推車的軌跡（從手推車的共移參考系視角）。這條條件下存在多種可能的情况。只畫其中一個。

(di) [2] Find the maximum value of θ that the particle will fly over the right side of the basket after at least one bounces off the side wall in the basket.

{di} [2] 找出粒子在籃子側壁至少反彈一次後會飛越籃子右側的最大 θ 值。

(dii) [2] Find the maximum value of θ that the particle will fly over the left side of the basket after at least two bounces off the side wall in the basket.

(dii) [2] 找出粒子在籃子側壁至少反彈兩次後會飛越籃子左側的最大 θ 值。

(e) [4] Find the values of θ such that the trolley will travel for a finite distance after the particle bounces off the trolley. Answer in terms of n , the number of hits on the side walls.

(e) [4] 找出使手推車在粒子反彈後能夠行進有限距離的 θ 值。答案以 n （側壁上的撞擊次數）表示。

(f) [3] For $\theta = 17^\circ$, find the total displacement of the trolley (in meter).

(f) [3] 當 $\theta = 17^\circ$ 時，找出手推車的總位移（以米為單位）。

Now consider a slightly different cart with unknown width L but with the same height $H = 1\text{m}$ and mass $M = 3\text{kg}$. In addition, the same particle flies into the basket at initial speed $v_0 = 1\text{ ms}^{-1}$ and angle $\theta = 0$, under a downward acceleration due to gravity $g = 9.8\text{ ms}^{-2}$.

現在考慮一個稍有不同手推車，其寬度 L 未知，但高度 $H = 1\text{m}$ 和質量 $M = 3\text{kg}$ 相同。此外，同樣的粒子以初始速度 $v_0 = 1\text{ ms}^{-1}$ 和角度 $\theta = 0$ 飛入籃子，受到向下的重力加速度 $g = 9.8\text{ ms}^{-2}$ 。

(g) [3] Suppose the particle will hit the bottom of the basket exactly once and hits at least once on one of the side walls. Find the maximum value of L such that the particle will barely flies away from the right side of the basket.

(g) [3] 假設粒子只碰到籃子底部一次，並至少碰到側壁一次。找出使粒子剛好飛出籃子右側的最大 L 值。

Answer:

(a) According to linear momentum conservation along x, we have

$$mv_0 \cos \theta = Mv_t + mv_{p,x} \rightarrow v_0 \cos \theta = 3v_t + v_{p,x} \quad - (1)$$

For energy conservation, since the vertical speed of the particle is constant, we only need to consider the energy due to the horizontal component. have

$$\frac{1}{2}mv_0^2 \cos^2 \theta = \frac{1}{2}mv_{p,x}^2 + \frac{1}{2}3mv_t^2 \quad - (2)$$

From equation (1) and (2), we have

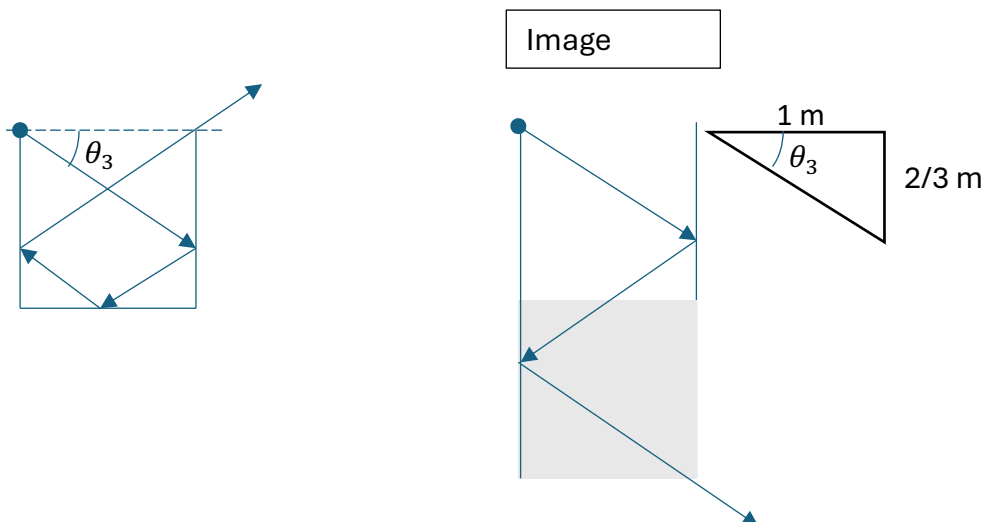
$$(a) \quad v_t = \frac{1}{2}v_0 \cos \theta$$

$$v_t = \frac{1}{2}v_0 \cos \theta$$

(b) From equation (1), we have $v_p = -\frac{1}{2}v_0 \cos \theta$

$$v'_p = v_p - v_t = -v_0 \cos \theta$$

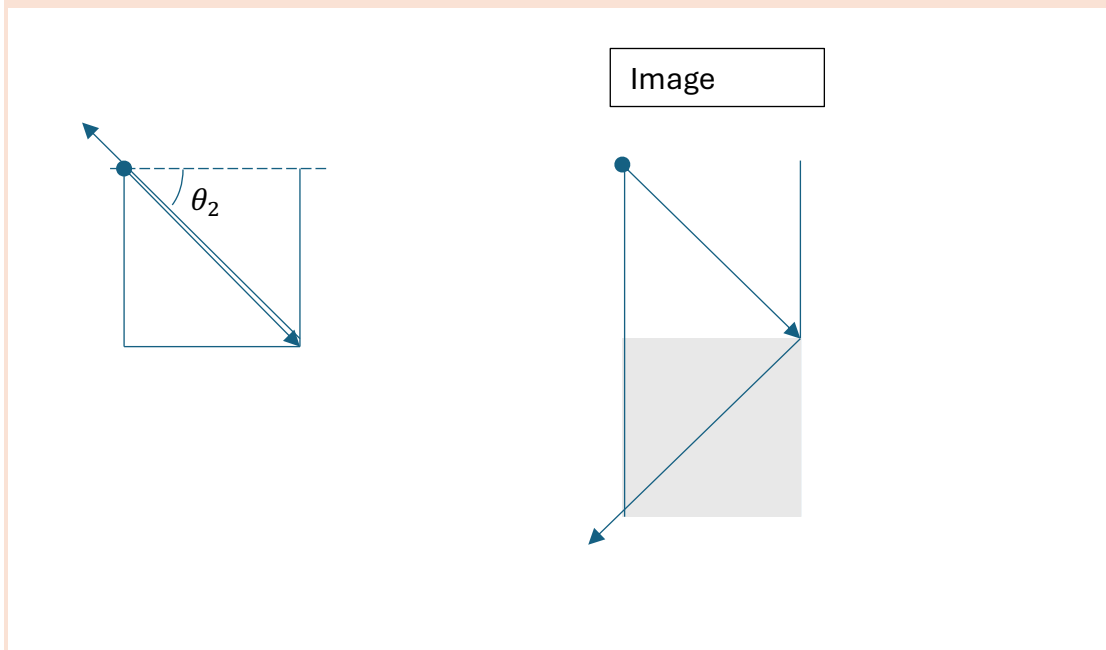
(c) Grading points: In the co-moving reference frame of the trolley, the particle reverses its horizontal velocity after every hit to the sides while keeping the vertical velocity unchanged. $\rightarrow$ the trajectory of the particle follows the law of reflection in optics.



Number of hit on the side walls	Exit side	Maximum angle	Minimum angle
0	R	$\pi/2$	$\arctan \frac{2}{1}$
1	L	$\arctan 2/1$	$\arctan 2/2$
2	R	$\arctan 2/2$	$\arctan 2/3$
3	L	$\arctan 2/3$	$\arctan 2/4$
...			
n	L: odd n R: even n	$\arctan \frac{2}{n}$	$\arctan \frac{2}{n+1}$

(d) (i) $\theta_2 < \arctan\left(\frac{2}{2}\right) = 45^\circ \text{ or } \frac{\pi}{4}$

(ii) $\theta_3 = \arctan\left(\frac{2}{3}\right) = 33.6^\circ$



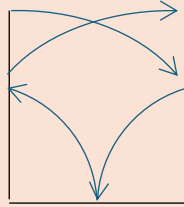
(e) As long as the particle leaves the basket from the right side, the trolley will stop moving once the particle hit the left side wall at the last time. Thus, the trolley will stop is the particle hits the wall even number of time. The range of the angles are

$$\frac{2}{2m+1} < \tan \theta < \frac{2}{2m} \text{ for } m = 1, 2, 3, \dots$$

(f) It hits the wall 6 times. The trolley only moves between (the first and second hit), (the third and fourth hit) and (the fifth and sixth hit) at the speed $\frac{1}{2} v_0 \cos \theta$. The time between the hit is $t = \frac{L}{v_0 \cos \theta}$. So the distance it travels is

$$3 \times \frac{1}{2} v_0 \cos \theta \times \frac{L}{v_0 \cos \theta} = 3/2 L = 1.5 \text{ m}$$

(g)



$$v_0 T = 3L \text{ (bounces 2 times } \rightarrow \text{ travelled 3 } L)$$

$$T = 2 \sqrt{\frac{2H}{g}}$$

$$L = \frac{2}{3} v_0 \sqrt{\frac{2H}{g}} = 0.30 \text{ m}$$

3 [20] Newton's law of gravitation holds only for two point objects. Nevertheless, for finite-sized objects with spherically symmetric mass distributions, the gravitational forces are still simple due to the following "Shell theorem":
The gravitational force by a thin (negligible thickness) spherical shell with uniform mass density exerting on a small test mass
(i) is the same as if all of its mass were concentrated at a point at its center, if the test mass is outside the shell, and
(ii) is zero (no net gravitational force), if the test mass is inside the shell.

3. 牛頓萬有引力定律僅適用於兩個點物體。對於球對稱質量分布的物體，基於”殼層定理”，其引力作用依然簡單。
殼層定理：

一個質量均勻分布的薄球殼層（厚度可忽略），其作用於一小試驗質量的引力有以下性質：

- (i) 如試驗質量位於殼層外，引力與假想所有殼層質量皆位於球心而給出的一樣。
- (ii) 如試驗質量位於殼層內，其受到的淨引力為零。

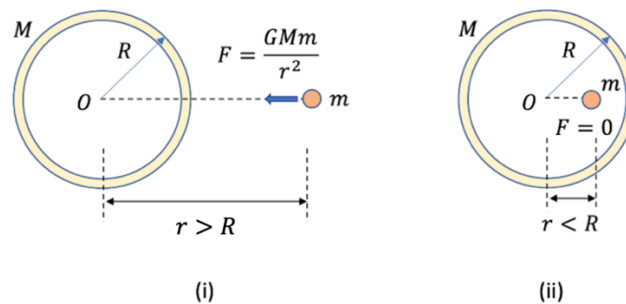


Figure: Shell theorem for (i) $r > R$ and (ii) $r < R$ respectively.

圖：分別為(i) $r > R$ 和(ii) $r < R$ 的殼層定理。

In this question, the Earth's density is assumed to be uniform, the Earth's self-rotation is ignored. All frictional forces can also be ignored.

You may find the following useful:

Earth's Mass M : $5.97 \times 10^{24} \text{ kg}$

Earth's Radius R : 6370 km

It is given that the shortest distance (along the Earth's surface) between Hong Kong and Moscow is $7.0 \times 10^3 \text{ km}$.

Suppose a straight narrow tunnel is dug through the Earth to connect Hong Kong and Moscow. The inner surface of the tunnel is smooth. The mass dug out is negligible.

A small object with mass m is released from rest at the end of the tunnel in Hong Kong to move inside the tunnel to Moscow.

本題假設地球密度均勻且忽略地球自轉，所有摩擦力亦可忽略。

以下數據可供參考：

地球質量 M ： 5.97×10^{24} 公斤

地球半徑 R ：6370 公里

已知香港與莫斯科沿地球表面的最短距離為 7.0×10^3 公里。

現若開鑿一條直線狹窄隧道貫穿地球連接香港與莫斯科，隧道內壁光滑，且挖掘出的質量可忽略不計。將質量為 m 的小物體從香港端隧道口以靜止狀態釋放，使其在隧道內運動到達莫斯科。

(a) [2] What is the magnitude of the gravitational force acting on the object when it is inside the tunnel at a distance r from the center of the Earth?

(a) [2] 當該物體位於隧道內距離地球中心 r 處時，作用於其上的引力大小為何？

$$\frac{GM_r m}{r^2} = \frac{Gm}{r^2} M \frac{r^3}{R^3} = \frac{GMm}{R^3} r$$

(b) [7] What is the average speed during its journey from Hong Kong to Moscow?

(b) [7] 從香港到莫斯科的路程中，其平均速度為何？

$$F(x) = -\frac{GMm}{R^3} r \frac{x}{r} = -\frac{GMm}{R^3} x$$

$$a(x) = -\frac{GM}{R^3} x$$

$$\omega = \sqrt{\frac{GM}{R^3}}$$

$$\frac{T}{2} = \frac{\pi}{\omega} = \pi \sqrt{\frac{R^3}{GM}} \approx 2530 \text{ s}$$

$$\theta = D/R = 7000/6370 \approx 1.10 \text{ rad}$$

$$L = 2R \sin \frac{\theta}{2} \approx 6650 \text{ km}$$

$$\frac{L}{T/2} = L \frac{\omega}{\pi} = \frac{L}{\pi} \sqrt{\frac{GM}{R^3}} = \frac{2}{\pi} R \sin \frac{\theta}{2} \sqrt{\frac{GM}{R^3}} = \frac{2}{\pi} \sqrt{\frac{GM}{R}} \sin \frac{\theta}{2} \approx 2.6 \text{ km/s}$$

(c) [9] If the object is launched with a downward speed of 9.00 km/s (along the tunnel), what are the minimum and maximum distances of it from the Earth's center during its journey, including that after it exits the tunnel at Moscow?

(c) [9] 若該物體以 9.00 公里/秒的向下速度（沿隧道方向）發射，其於路程中（包括從莫斯科的隧道出口離開後）距離地球中心的最小與最大距離分別為何？

$$r_{\min} = R \cos \frac{\theta}{2} = 5400 \text{ km}$$

$$\frac{1}{2} m v_0^2 - \frac{GMm}{R} = \frac{1}{2} m v^2 - \frac{GMm}{r_{\max}}$$

$$m v r_{\max} = m v_0 R \sin \left(90^\circ - \frac{\theta}{2} \right) = m v_0 R \cos \frac{\theta}{2} = m v_0 r_{\min}$$

$$v = v_0 \frac{r_{\min}}{r_{\max}}$$

$$v_0^2 - \frac{2GM}{R} = v^2 - \frac{2GM}{r_{\max}} = v_0^2 \frac{r_{\min}^2}{r_{\max}^2} - \frac{2GM}{r_{\max}}$$

$$\left(\frac{2GM}{R} - v_0^2\right)r_{\max}^2 - 2GM r_{\max} + v_0^2 r_{\min}^2 = 0$$

$$(2GMR - v_0^2 R^2) \left(\frac{r_{\max}}{R}\right)^2 - 2GMR \frac{r_{\max}}{R} + v_0^2 r_{\min}^2 = 0$$

$$\left(1 - \frac{v_0^2 R}{2GM}\right) \left(\frac{r_{\max}}{R}\right)^2 - \frac{r_{\max}}{R} + \frac{v_0^2 R}{2GM} \left(\frac{r_{\min}}{R}\right)^2 = 0$$

$$(1 - \alpha) \left(\frac{r_{\max}}{R}\right)^2 - \frac{r_{\max}}{R} + \alpha \left(\frac{r_{\min}}{R}\right)^2 = 0$$

where

$$\alpha = \frac{v_0^2}{2GM/R}$$

$$r_{\max} = \frac{1 + \sqrt{1 - 4\alpha(1 - \alpha) \left(\frac{r_{\min}}{R}\right)^2}}{2(1 - \alpha)} R = \frac{1 + \sqrt{1 - 4\alpha(1 - \alpha) \cos^2 \frac{\theta}{2}}}{2(1 - \alpha)} R$$

$$r_{\max} \approx 14000 \text{ km}$$

(d) [2] If another tunnel is constructed similarly to connect Hong Kong to the opposite point on the Earth (i.e., the tunnel passes through the Earth's center). What is the gravitational potential energy when the object is inside the tunnel at a distance r from the center? Take infinity as the zero level of the gravitational potential energy. Express your answer in terms of M , m , R , and r .

(d) [2] 若以類似方式建造另一條隧道連接香港與地球正對面的一點（即隧道通過地球中心），當物體位於隧道內距離地心 r 處時，其引力勢能為何？取無窮遠處為引力勢能的零參考點。答案以 M 、 m 、 R 及 r 表示。

$$-\frac{GMm}{R} - U(r) = \frac{1}{2} k [R^2 - r^2] = \frac{1}{2} \frac{GMm}{R^3} [R^2 - r^2]$$

$$U(r) = -\frac{GMm}{R} - \frac{1}{2} \frac{GMm}{R^3} [R^2 - r^2] = -\frac{1}{2} \frac{GMm}{R^3} (3R^2 - r^2)$$

~ END of PAPER ~